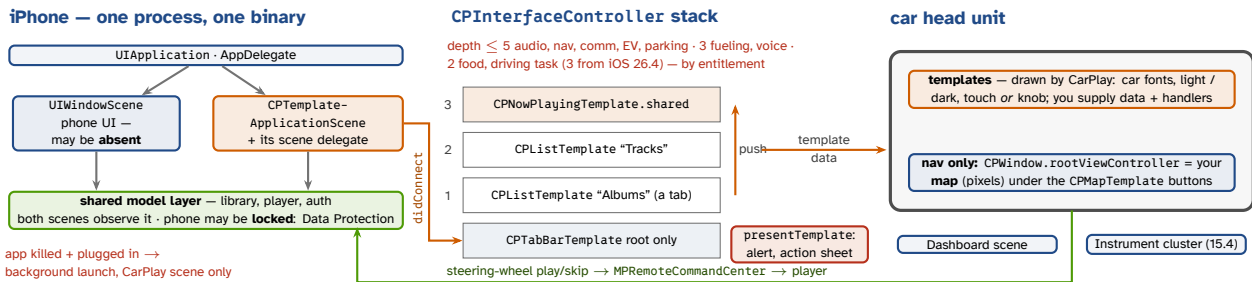


In one line: A CarPlay app is **not a separate target**: it is a second **scene** (CPTemplateApplicationScene) of your iOS app, same process. You never draw its UI — you hand **templates** to a CPInterfaceController and the **car renders them**. Apple grants one **category entitlement**; the category decides which templates you may use and how **deep** the stack goes. Only **navigation** apps draw pixels (the map, in a CPWindow).



How it works

- Entitlement** per category, requested from Apple; a Boolean key in .entitlements, matching App ID + profile. A template outside your category → **runtime exception**.
- Scene manifest:** UISceneConfigurations gets a CPTemplateApplicationSceneSessionRoleApplication entry (class CPTemplateApplicationScene + your delegate). No extension, no new target.
- Connect:** templateApplicationScene(_:didConnect:) hands you the CPInterfaceController: keep it, setRootTemplate. Nav apps get ...didConnect:to: with a CPWindow. Disconnect → drop both.
- Stack:** push / pop / pop(to:); presentTemplate shows one modal. CPTabBarTemplate is root only, each tab its own stack.
- Templates are data:** text, CPIImageSet (light/dark), handlers. Update in place (updateSections(_)).
- The car sets limits:** maximumItemCount, maximumTabCount, CPSessionConfiguration.limitedUserInterfaces (.keyboard, .lists while driving); a grid shows 8 buttons.

Categories — key com.apple.developer.carplay-...

category	key suffix	depth	own templates · rule
audio	audio	5	Now Playing
communication	communication	5	Contact
navigation	maps	5	Map, Search, Voice
EV charging, parking	charging, parking	5	Point of Interest, Information
quick food ordering	quick-ordering	2 (3 [†])	Point of Interest, Information
fueling (iOS 16)	—	3	
driving task (iOS 16)	driving-task	2 (3 [†])	
public safety	—	—	dispatch, routing
voice conv. (26.4)	voice-based-conversation (unverified)	3	audio session only while voice is in use
video (iOS 27)	—	—	parked only, AirPlay video

[†] iOS 26.4+. List, grid, tab, alert, action sheet: all. Information: not audio. EV + fueling combine, audio + video too. Old MPPlayableContentManager: deprecated (14).

Example — audio app, scene delegate

```
func templateApplicationScene(_ s: CPTemplateApplicationScene,
                             didConnect ui: CPInterfaceController) {
    self.ui = ui // keep it
    let max = CPListTemplate.maximumItemCount // the car's limit
    let rows = Library.shared.albums.prefix(max).map { a in
        let row = CPListItem(text: a.title, detailText: a.artist)
        row.handler = { _, done in
            Player.shared.play(a) // shared model
            ui.pushTemplate(CPNowPlayingTemplate.shared,
                           animated: true, completion: nil)
            done() // or it spins forever
        }
        return row
    }
    let list = CPListTemplate(title: "Albums",
                             sections: [CPListSection(items: rows)])
    ui.setRootTemplate(list, animated: true, completion: nil)
}
```

Testing

Xcode Simulator **I/O** → **External Displays** → **CarPlay** (800×480 @2x): no **locked** iPhone, **Siri** or **audio**. **CarPlay Simulator** (Mac app, Additional Tools for Xcode) drives a **real iPhone** over a cable.

Traps

- ! Model built in the phone SceneDelegate: CarPlay may launch you with *no* phone scene.
- ! Locked phone: .complete files and WhenUnlocked keychain items fail — use completeUntilFirstUserAuthentication / AfterFirstUnlock.
- ! Hard-coded row counts: read the limits at runtime; they shrink while driving.
- ! "Custom CarPlay UI": only the nav map. No video (bar parked video apps), games or free text.

Per category

- Audio:** browse → CPNowPlayingTemplate.shared, a singleton fed by MPNowPlayingInfoCenter; its buttons and the steering wheel go through MPRemoteCommandCenter — the lock-screen plumbing.
- Communication:** SiriKit (INSendMessageIntent, INSearchForMessagesIntent, INStartCallIntent) + CallKit; CPContactTemplate, CPMessageListItem. Siri dictates — no keyboard.
- Navigation:** CPMapTemplate (buttons, trips, maneuvers via CPNavigationSession) over your map; CPSearchTemplate; Dashboard (CPSupportsDashboardNavigationScene) + instrument-cluster scenes.
- EV / parking / food:** CPPointOfInterestTemplate (map + picker), CPIInformationTemplate (items + ≤ 3 CPTextButtons).

New in iOS 26 · 26.4 · 27

- 26:** .systemSmall widgets + Live Activities (.small) show on the Dashboard, no CarPlay app needed; opt out: .disfavoredLocations([.carPlay], for: [.systemSmall]).
- 26.4:** voice conversational apps; food/driving-task depth 3.
- 27:** **video** apps — parked, in cars with video-in-car; AirPlay video; audio-only fallback (supportsVideoPlayback). CPVoiceControlTemplate for all.
- CarPlay Ultra** (every car display): automaker feature; templates restyle.

Remember

"The entitlement picks the templates, templates are data, the car draws — only maps get pixels."